

High-chance portions — geo-physical

Combined original study articles for this topic's highest-probability PYQ themes.

Indian Monsoon & Climatology Essentials

~44% · Monsoon mechanism · Jet streams · El Niño/IOD · Cyclones

ITC Z, jet streams, EL Niño links and monsoon mechanisms — yearly Prelims staple.

Why UPSC asks this

Monsoon integrates atmosphere dynamics with Indian livelihood — perfect for mechanism + statement questions.

Mechanism in exam language

Explain monsoon as seasonal reversal of winds driven by differential heating of land and ocean, modulated by the Intertropical Convergence Zone's northward shift, orography, and upper-air jets. Know onset/progress/withdrawal as map stories, not only definitions.

Teleconnections

El Niño–Southern Oscillation and Indian Ocean Dipole appear in recent papers. Treat correlations as probabilistic. Tropical cyclones in Bay of Bengal/Arabian Sea: conditions (warm SST, Coriolis, low shear) — link to disaster themes.

Pressure belts & world climate bridge

Hadley cells, horse latitudes, westerlies — still asked in physical geography blocks feeding monsoon understanding.

Must remember

- Southwest monsoon June–Sep; retreat/NE monsoon affecting Tamil Nadu coast
- Tibetan heating + Mascarene High + Somali jet ingredients in standard models
- El Niño often correlated with weak monsoon — not iron law

Geomorphology & Oceanography Quick Stack

~40% · Plate tectonics · Landforms · Ocean currents · Coral reefs

Plate tectonics, landforms by agents, currents, coral reefs and EEZ basics.

Why UPSC asks this

Diagrams-in-words and location–process matching recur in GS Prelims.

Solid earth & landforms

Interior layers, earthquakes/volcanoes at plate edges, weathering vs erosion, fluvial/karst/glacial/aeolian landforms. For India, know Himalayan vs Peninsular structural contrast.

Oceans

Currents (Gulf Stream, Humboldt, Labrador, Kuroshio — exemplar set), tides, EEZ (200 nm concept), coral reef conditions (temperature, salinity, clear water) and bleaching drivers.

Must remember

- Convergent/divergent/transform boundaries
- Himalayan orogeny = continental collision
- Warm vs cold currents impact climate/fisheries